# ABINESH SUBRAMANI

**Senior Backend Engineer · Java · Distributed Systems · Cloud-Native · Agentic AI**

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## SUMMARY

Backend engineer with 8+ years building Java/Spring Boot microservices and Spark pipelines on Azure, Kubernetes and Docker for fintech reconciliation and identity platforms. Owns a reconciliation platform at Societe Generale that processes 1.5B+ records a month. Founder and sole engineer of RefinePoint, a live multi-tenant reconciliation SaaS with agentic AI features.

## EXPERIENCE

**Societe Generale Global Solutions Centre** Nov 2022 – Present

*Specialist Software Engineer* *Bengaluru*

- Led design and delivery of a reconciliation platform (Java 17, Spring Boot, Apache Spark) processing 1.5B+ financial records a month at 99.9% availability; owned it from architecture to production rollout and post-release monitoring.
- Re-engineered Spark batch and streaming jobs, raising transformation throughput by 60% and cutting processing from hours to minutes; multi-key matching of 7M+ records now completes in about 40 minutes.
- Built Orbiter, an in-house orchestrator that replaced Apache Airflow: runs configurable multi-step pipelines, submitting each Spark job to Kubernetes as a SparkApplication (Spark Operator) when the previous step completes.
- Built 10+ Spring Boot microservices for financial data onboarding, file routing and reconciliation.
- Designed configuration-driven REST APIs that onboard new data sources without code changes, cutting onboarding time by 40%.
- Built event-driven flows on Azure Event Hub that keep end-to-end latency under 3 seconds for critical data; ran services on Kubernetes with autoscaling and zero-downtime rolling deployments.
- **Pinnacle Performer Award 2024** for owning the Spark reconciliation pipelines; **Team of the Year 2024** for the front-office/back-office (FOBO) reconciliation platform, building its core matching and break management.

**Publicis Sapient** Jul 2022 – Nov 2022

*Software Engineer*

- Built Spring Boot microservices, CI/CD pipelines (Jenkins, GitLab CI, Azure DevOps) and Docker images for Kubernetes deployment.

**Tata Consultancy Services** Jul 2018 – Jul 2022

*IT Analyst*

- Built backend services and REST APIs (Java, Spring Boot) for a multi-tier authentication and access platform serving 300,000+ users, integrating Okta, OpenLDAP and CA Directory.
- Developed integration modules for Ping Federate, Okta and CA Identity Manager covering single sign-on, role management and access-request workflows.
- Migrated legacy identity applications to Spring Boot microservices on Azure Kubernetes Service (AKS), with Azure Service Bus for event-driven communication.

## PROJECT

**RefinePoint – reconciliation SaaS for finance teams** 2026 – Present

*Founder and sole engineer · refinepoint.com* *Java 21 · Spring Boot 3 · Kafka · React 19*

- Built a multi-tenant SaaS end to end: 14 Spring Boot microservices behind Spring Cloud Gateway, Kafka events, PostgreSQL schema-per-service (Flyway), Keycloak OIDC/SSO and a React + TypeScript UI; live with Paddle billing.
- Wrote the matching engine: multi-pass two-, three- and four-way matching, with data preparation pushed down to DuckDB SQL so it scales with volume; a custom expression language and plugin API let customers define their own rules.
- Built an AI service on a provider-agnostic LLM layer (Anthropic Claude, OpenAI-compatible and local models) that drafts reconciliations from plain English, suggests mappings, triages exceptions, flags unusual runs (median/MAD) and reads statements from PDFs and images; answers are validated against real workspace data, behind prompt-injection guards and rate limits.
- Built two chat assistants: a RAG help bot that ranks 73 help articles with BM25, answers only from the best sections with source links and skips the model when nothing matches; and a results bot that turns “unmatched items over 10,000 from vendor X” into a validated filter, so the model never writes SQL.
- Built agentic AI: a rule-builder agent that writes rules in a custom DSL by calling tools (validate, test on sample records, inspect lookup tables) for up to 8 steps, accepting an answer only once it validates; and an MCP server with 22 tools that lets external agents (Claude Desktop, Claude Code) design and run reconciliations under the user’s own permissions.

## SKILLS

**Languages & backend:** Java (8–21), SQL, TypeScript, Spring Boot, Spring Cloud Gateway, Kafka, Azure Event Hub, Service Bus

**Data:** Apache Spark (Spark on Kubernetes), PostgreSQL, MySQL, DuckDB, Apache Airflow, Quartz

**AI:** Agentic AI (tool-calling agents, MCP), Spring AI, LLM integration (Claude, OpenAI-compatible APIs), RAG, chat assistants, structured output, prompt-injection defence

**Cloud & DevOps:** Azure, Kubernetes (AKS), Docker, Jenkins, Azure DevOps, GitLab CI, Terraform, ELK, Prometheus, Grafana, App Insights

**Identity & testing:** OAuth 2.0, OIDC, SAML, Keycloak, Okta, Ping Federate, OpenLDAP, JUnit, Mockito, SonarQube

## EDUCATION & CERTIFICATION

**B.E. Electrical and Electronics Engineering**, RMK Engineering College 2014–2018

**Microsoft Certified: Azure Administrator Associate** (AZ-104)